

Synthetic Satellite Segmentation

Final Rural-to-Urban Generalization Report

Abstract. This study examined whether lightweight synthetic image augmentation or rare-class sampling can help a satellite-image segmentation model transfer from Rural to Urban environments. A compact U-Net was trained only on 1,366 Rural LoveDA image-mask pairs. Three fixed training interventions were compared across seeds 42, 7, and 123: a real-only baseline, 10% mild brightness copies, and Barren-focused oversampling. Frozen checkpoints were evaluated on 992 Rural validation images and 677 unseen Urban validation images. The real-only baseline fell from 24.64% Rural mIoU to 21.44% Urban mIoU. Brightness augmentation achieved 23.16% mean Urban mIoU, a +1.73 percentage-point average gain over the baseline, but with high variation across seeds. Barren-focused sampling reached 22.20% mean Urban mIoU (+0.77 points) and raised Urban Barren IoU from 0.00% to 5.44%. The results support Barren-focused sampling as a targeted rare-class intervention and identify brightness augmentation as promising but not yet conclusive for cross-environment transfer.

1. Research Question

Can lightweight synthetic image augmentation and rare-class sampling improve a satellite segmentation model's ability to generalize from Rural to Urban environments?

Plain-language version

The model learned from countryside satellite images only. We then showed it city images that it had never seen during training. The question is whether simple artificial changes to the training data, or extra attention to a rare land type, make the model more useful in that new environment.

2. Data and Experimental Design

Component	Design choice
Training data	1,366 labeled LoveDA Train/Rural images. Urban images were never used to train or choose a checkpoint.
Evaluation data	992 Val/Rural images for in-domain performance and 677 Val/Urban images for unseen-environment transfer.
Model	Compact U-Net, seven land-cover classes, 512 x 512 inputs, CrossEntropyLoss with No Data ignored.
Controls	Same architecture, learning rate, batch size, epochs, and Rural training budget for all compared experiments.
Repeated runs	Three random seeds: 42, 7, and 123. Results are reported as the mean across seeds, with variation shown.

Interventions

Experiment	Training intervention	Purpose
Real-only baseline	Original Rural images only.	Reference result.
10% brightness	10% of the final training mix was mildly darker or brighter copies.	Make the model less dependent on one lighting condition.
Barren-focused sampling	Rural images containing Barren land were sampled more often.	Help the model learn a rare class that the baseline missed.

3. Results

Overall Rural-to-Urban Performance

Experiment	Rural mIoU	Urban mIoU	Urban change vs baseline	Urban pixel accuracy
Real-only baseline	24.64% +/- 0.18%	21.44% +/- 1.02%	Reference	45.28%
+10% brightness	23.71% +/- 0.65%	23.16% +/- 3.43%	+1.73 points	46.54%
Barren-focused sampling	24.60% +/- 0.40%	22.20% +/- 1.13%	+0.77 points	45.04%

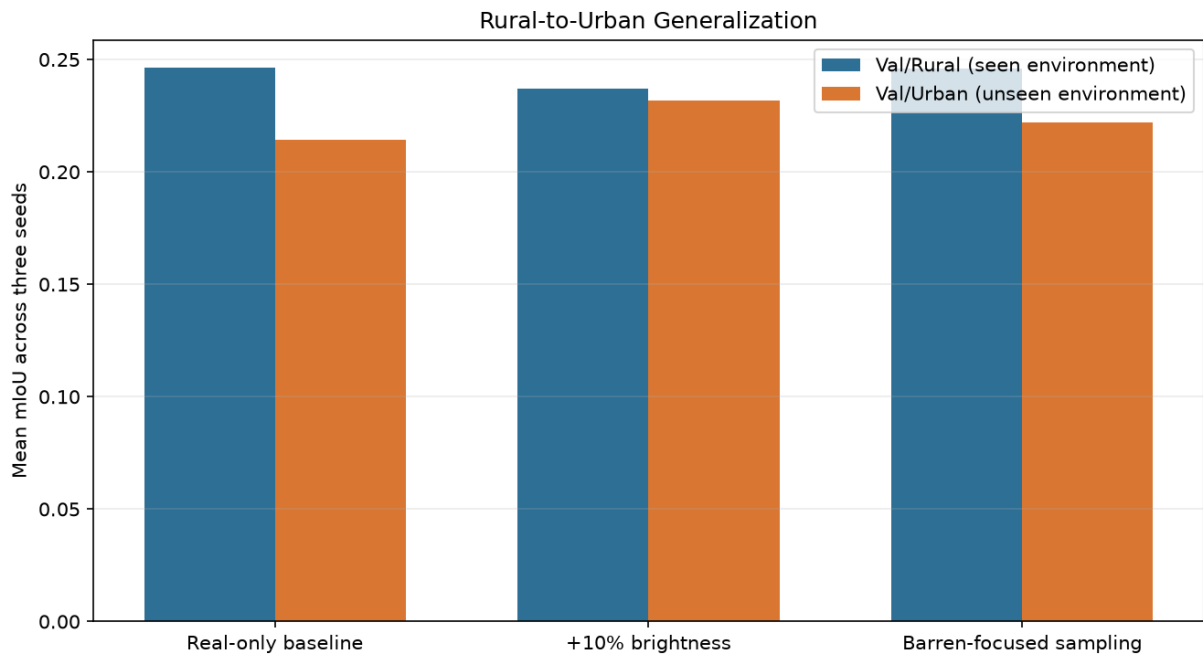


Figure 1. Mean mIoU across the three seeds. The gap between the Rural and Urban bars shows the cost of transferring to an unseen environment.

Key findings

- 1. A real domain gap exists.** The real-only Rural-trained model lost 3.21 mIoU points when moved from Rural validation to Urban validation. This confirms that Urban imagery was a more difficult, unseen environment for the model.
- 2. Brightness is encouraging but uncertain.** Its average Urban mIoU was 1.73 points higher than the real-only baseline and its average Urban pixel accuracy was also higher. However, the 3.43-point standard deviation shows the result differed substantially by seed. It is a promising signal, not proof of a reliable improvement.
- 3. Barren-focused sampling solved a specific failure.** The baseline had 0.00% Urban Barren IoU. The Barren-focused method reached 5.44% average Urban Barren IoU and a modest +0.77-point Urban mIoU gain.

Urban Per-class IoU

Class	Baseline	+10% brightness	Barren-focused
Background	30.52%	30.72%	30.64%
Building	12.00%	18.28%	14.96%
Road	3.50%	8.30%	11.86%

Class	Baseline	+10% brightness	Barren-focused
Water	45.83%	48.40%	45.73%
Barren	0.00%	0.00%	5.44%
Forest	22.16%	19.84%	11.81%
Agriculture	36.05%	36.61%	34.97%

4. Visual Evidence and Interpretation

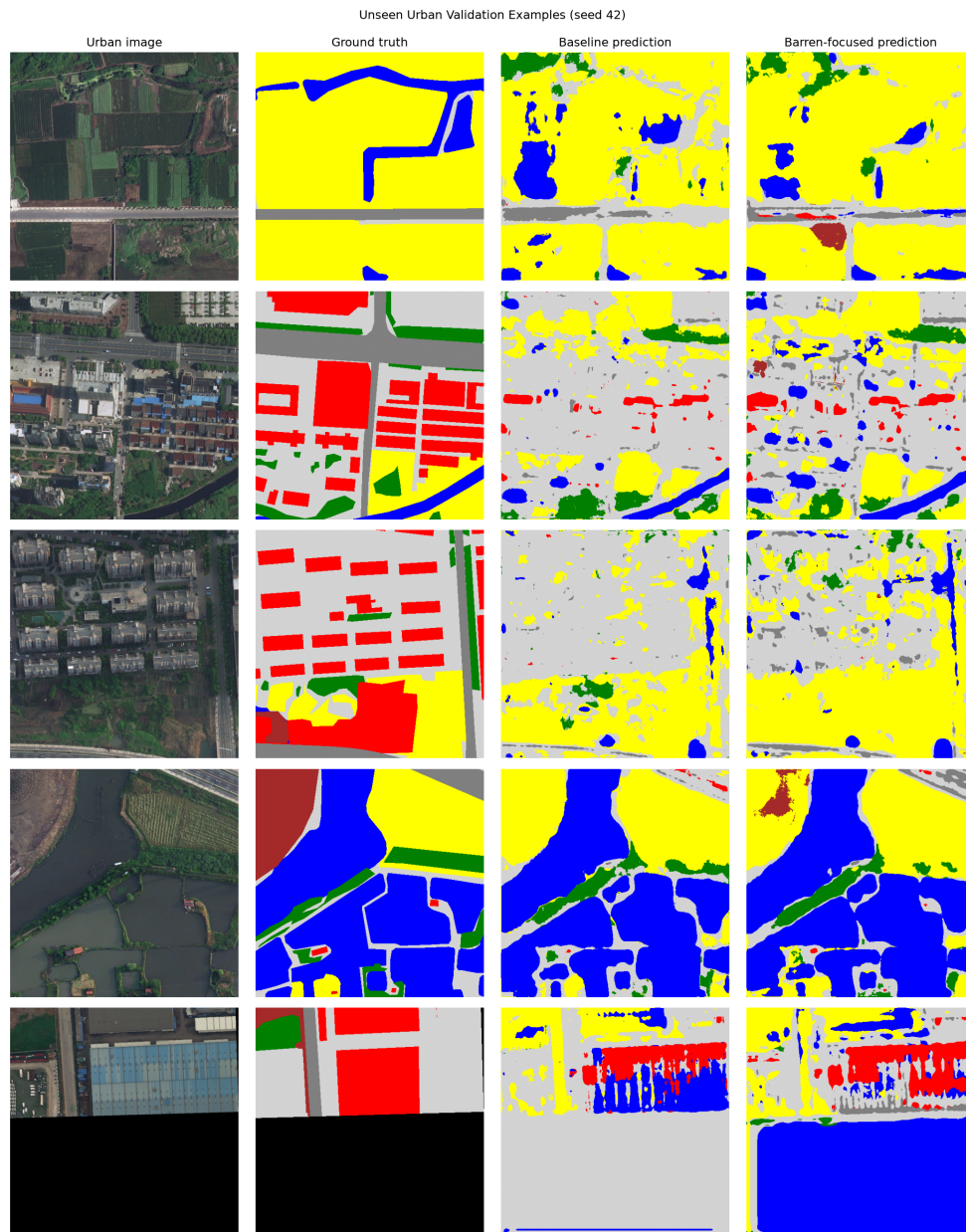


Figure 2. Five unseen Urban validation examples. Columns show the image, ground truth, real-only baseline prediction, and Barren-focused prediction. The colors are fixed land-cover labels.

What the examples show

The models identify some broad structures, especially water and large road regions, but make fragmented predictions in dense urban scenes. Building boundaries and narrow roads remain difficult. The Barren-focused model adds occasional Barren predictions that the baseline does not produce, but it can also trade off precision in other classes. The visual evidence agrees with the numerical results: there is meaningful transfer, but the Rural-trained model is still far from a complete Urban solution.

5. Conclusion

Answer to the research question: Lightweight interventions can help some aspects of Rural-to-Urban transfer, but not in a universal or guaranteed way. Mild brightness augmentation produced the highest average Urban score, yet its run-to-run variation is too large for a strong claim. Barren-focused sampling consistently gave the model nonzero ability to recognize an otherwise missed rare class and slightly improved mean Urban mIoU. Therefore, targeted interventions are more credible than generic augmentation, while brightness deserves further confirmation.

Limitations

The study evaluates one dataset benchmark, one compact U-Net, and three random seeds. Urban validation labels were used only after training, but the result should be described as Rural-to-Urban transfer within LoveDA rather than proof of global geographic generalization. Brightness has high seed variation, so more independent runs would be needed before treating its average gain as stable.

Recommended next study

The current study is complete. If extended, the most useful next experiment would be a fourth and fifth matched seed for the real-only and brightness conditions, evaluated with this same Rural-to-Urban protocol. That would test whether the brightness improvement is repeatable rather than a result driven by the particularly strong seed-123 run.

References

- [1] Wang, J., et al. LoveDA: A Remote Sensing Land-Cover Dataset for Domain Adaptive Semantic Segmentation. arXiv:2110.08733, 2021. <https://arxiv.org/abs/2110.08733>
- [2] Ronneberger, O., Fischer, P., and Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. arXiv:1505.04597, 2015. <https://arxiv.org/abs/1505.04597>
- [3] Project evaluation artifacts: evaluation/rural_to_urban/results.json, comparison.md, rural_urban_miou_chart.png, and urban_validation_examples_seed42.png.